

Temperature- and Humidity-Aware Low-Cost Bridge Vibration Anomaly Detection

Team SENTRY

01 The Problem

1. Sudden collapses



Source: Safety Journal

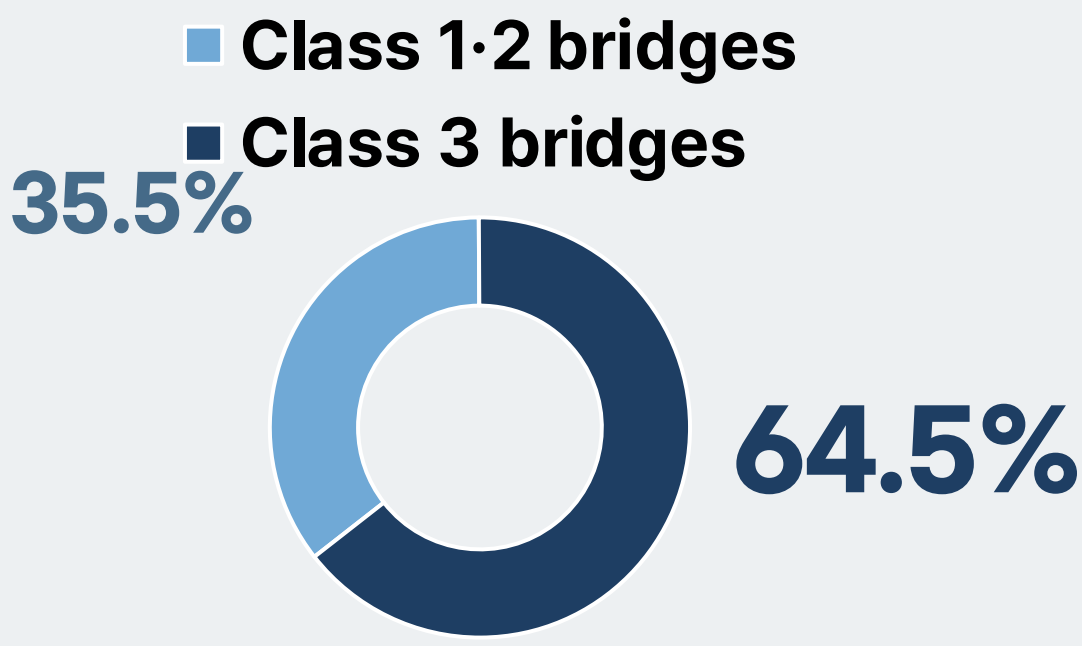
Jeongja Bridge collapse (2023)

Source: Yonhap News

Boritdol Bridge collapse (Jul 2026)

- Collapsed before the next inspection, even after inspection and repair
- Condition changes between checks are hard to track

2. Monitoring blind spot



Source: Korea Authority of Land & Infrastructure Safety, 2025

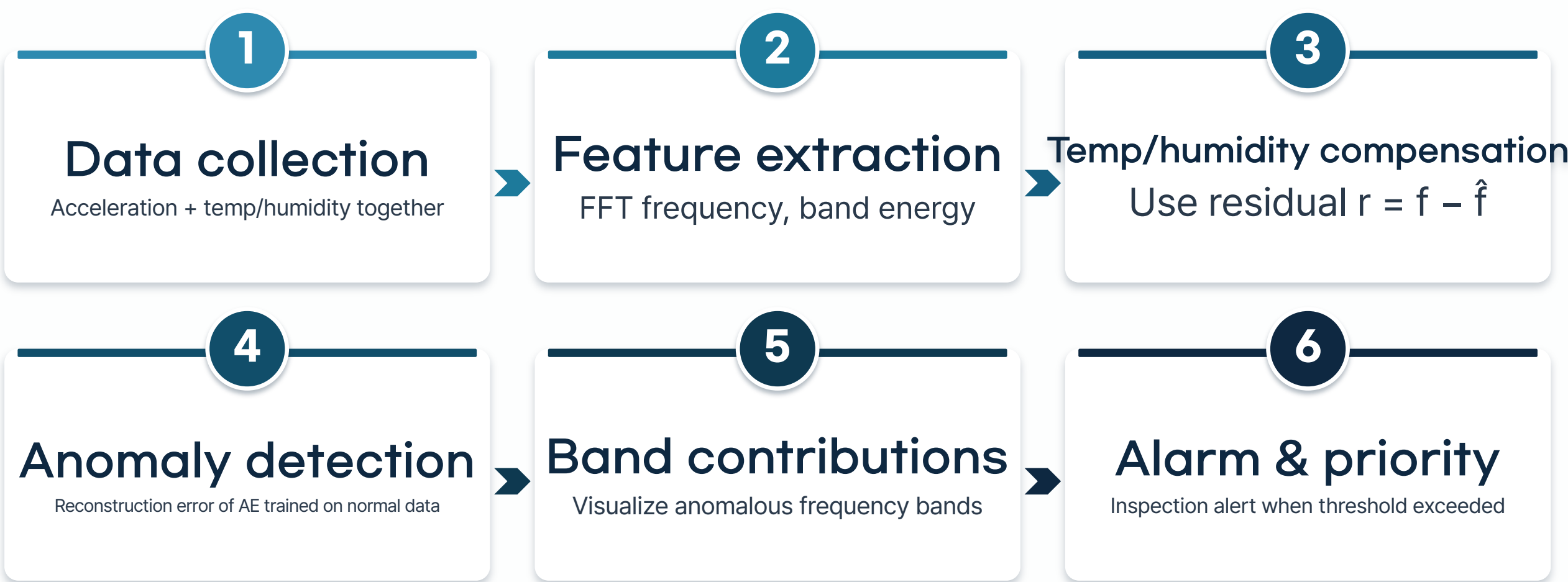
- Class 3 (small/medium): 64.5% of Korea's 37,543 bridges
- Precision diagnosis only every 4–6 years

3. No low-cost technology



- Existing SHM costs tens of millions of won — mostly large bridges
- No low-cost screening before precision diagnosis

02 Solution & How It Works



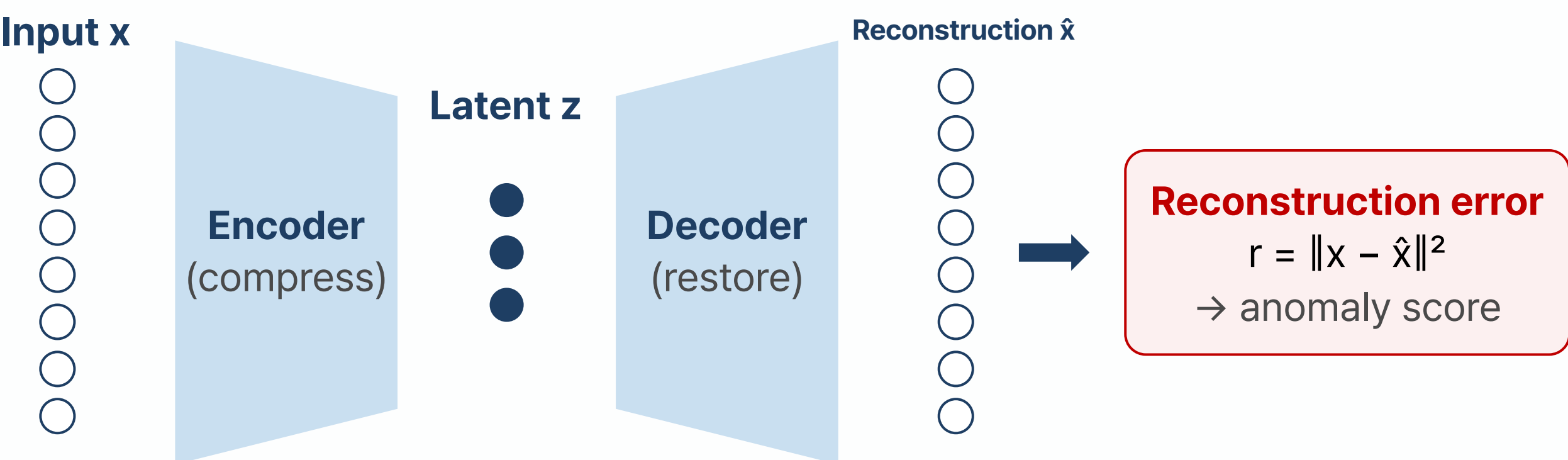
① Environmental compensation — temp/humidity steps 1-3

- Learn the temp/humidity–vibration feature relation by regression, remove environmental variation → use residuals only



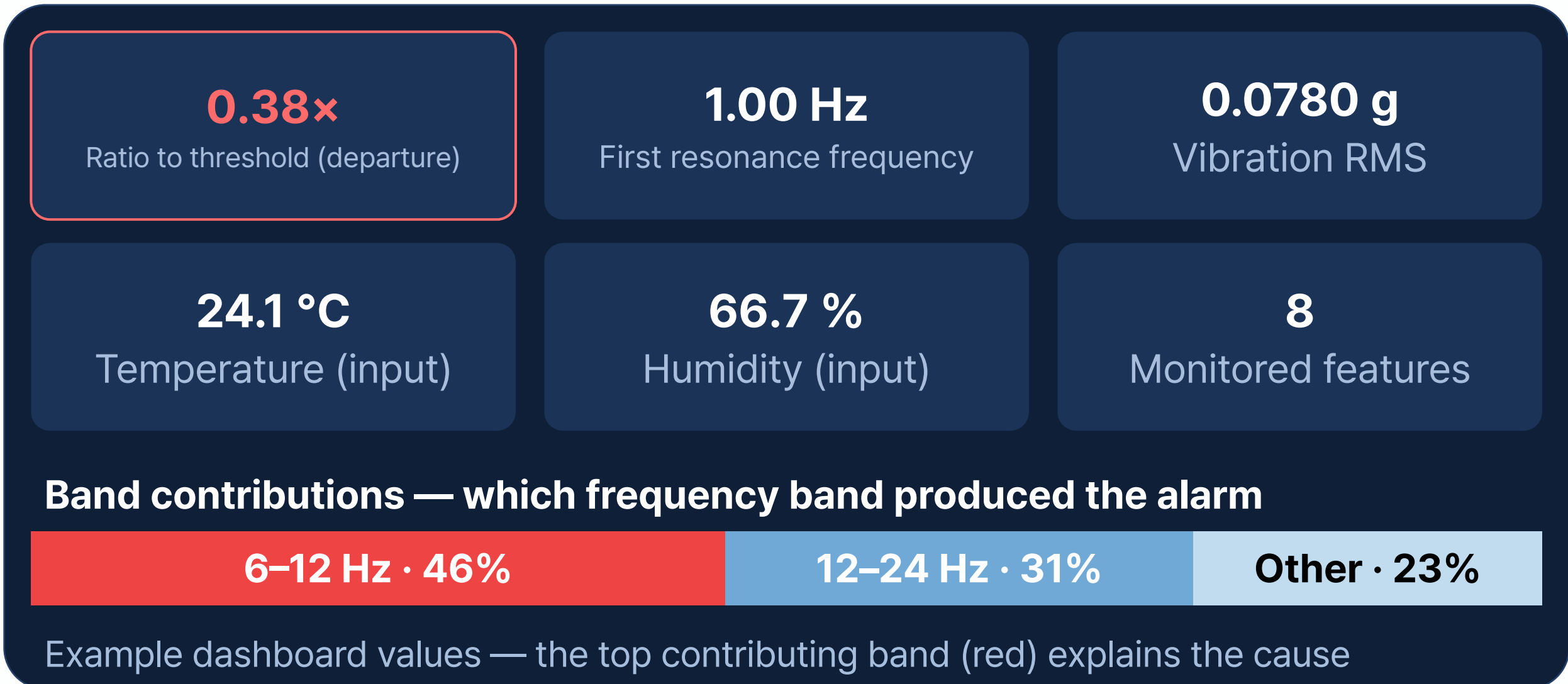
② Unsupervised detection — Autoencoder (8-3-8) step 4

- Trained only on normal vibration, no damage data → reconstruction error rises with damage
- Suited to bridges in service, which have no damage labels

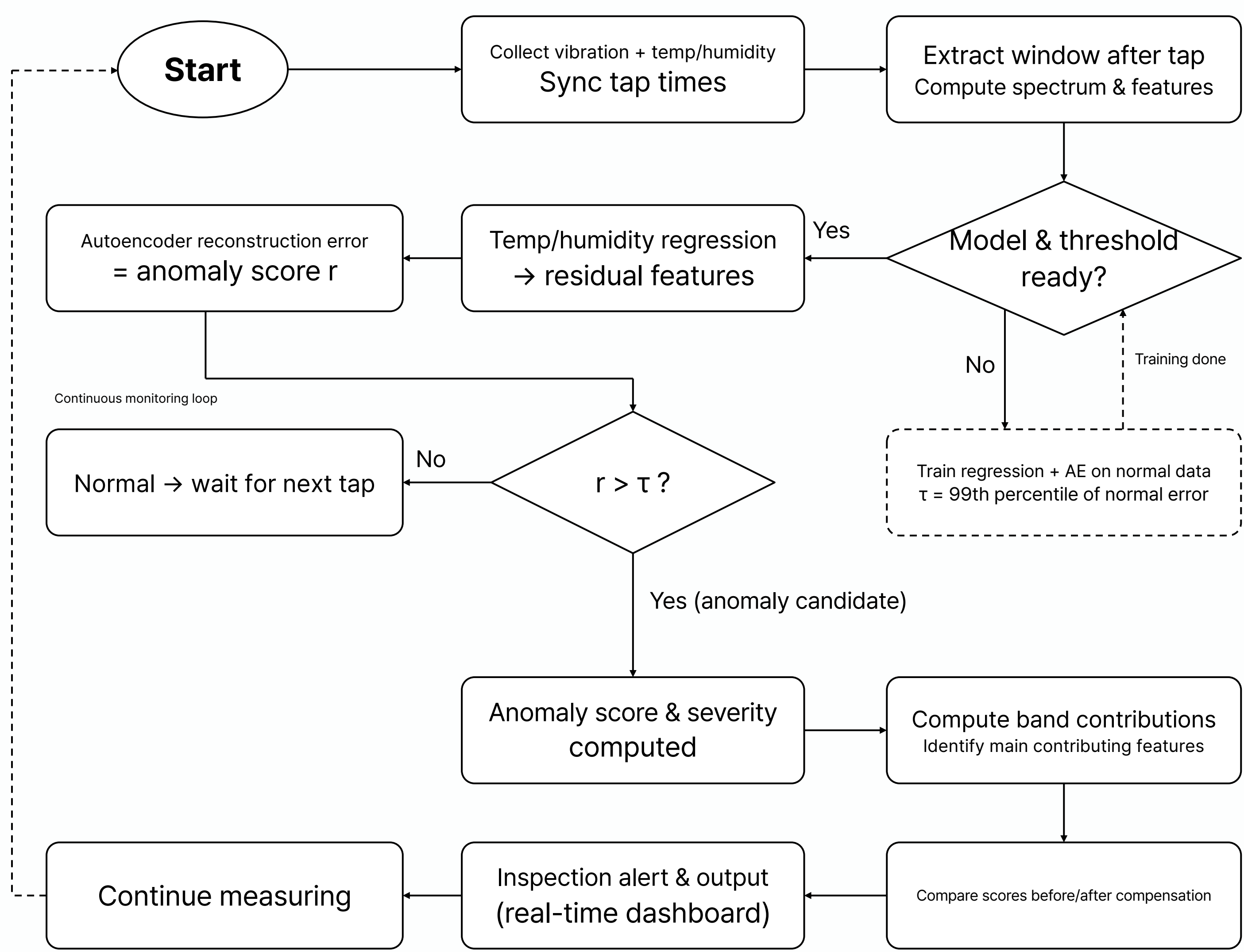


③ Explainable alarms — threshold & contributions steps 5-6

- Threshold set automatically at the 99th percentile of normal error
- Shows the frequency band behind the alarm → explains why it fired, guides inspection
- Ratio to threshold quantifies departure from normal → sets inspection priority



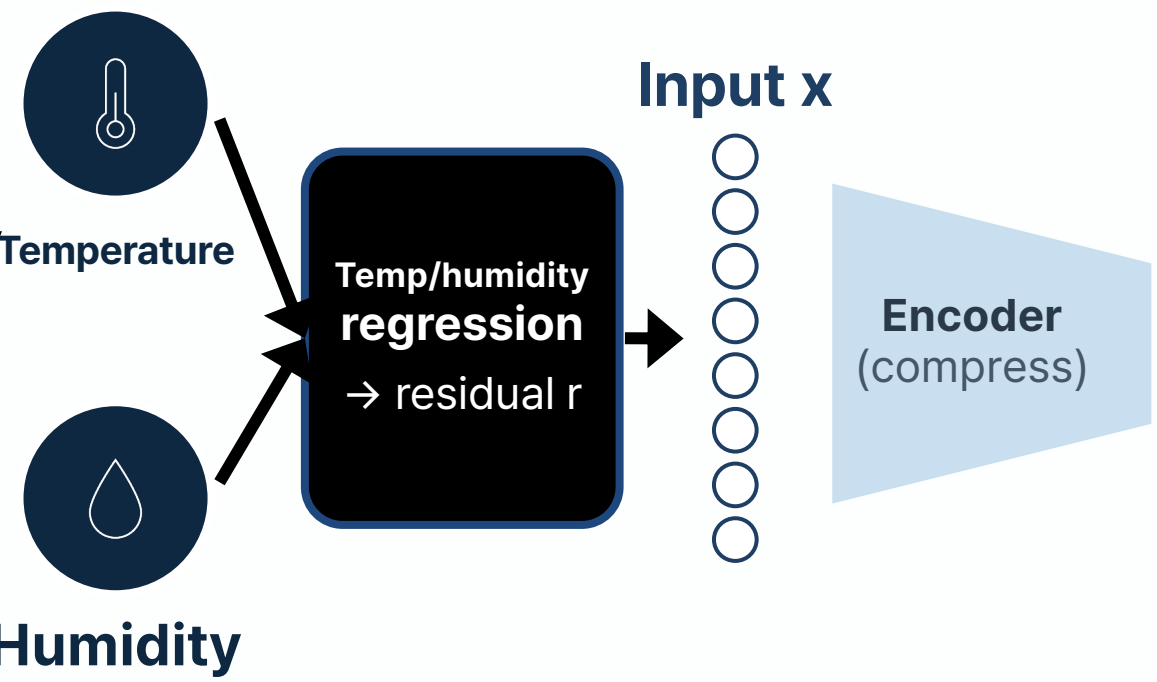
03 Flowchart



04 Differences from Similar Work

1) Temperature-based compensation research

- Compensates temperature only; ignores humidity
- Compensation results not presented as grounds for alarms
- Difference
- Temp/humidity residuals separate environmental effects
- Shows scores before/after compensation as alarm grounds



2) ML-based anomaly detection research

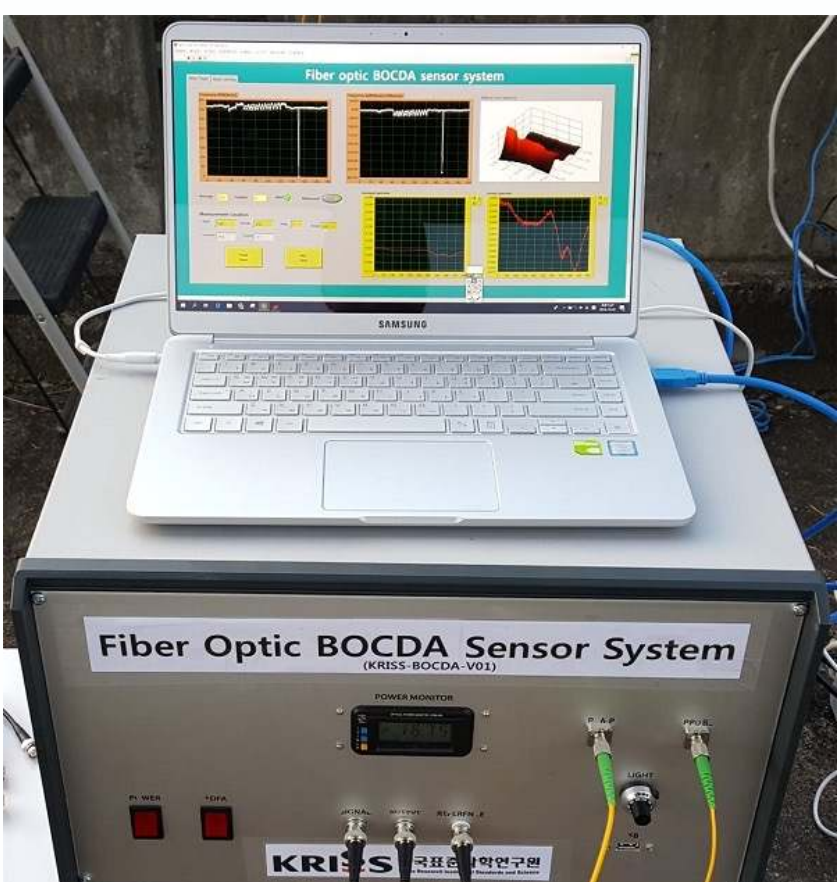
- Outputs only an anomaly score or threshold exceedance
- Interpreting the cause after an alarm relies on experts
- Difference
- Band contributions show what is abnormal and why
- Ratio to threshold quantifies departure and sets priority

Bridge	Ratio	Decision
1 Bridge A	65.0×	Priority 1
2 Bridge B	22.6×	Priority 2
3 Bridge C	5.8×	Priority 3
4 Bridge D	0.7×	Normal

<Inspection priority (hypothetical example)>

3) Fiber-optic (BOCDA) measurement

- Costly equipment and dedicated infrastructure
- Hard to run continuously without specialist analysts
- Difference
- Cheap general-purpose parts screen many bridges
- Complements, not replaces — picks bridges to measure



Source: KRIS

<BOCDA fiber-optic measurement equipment>

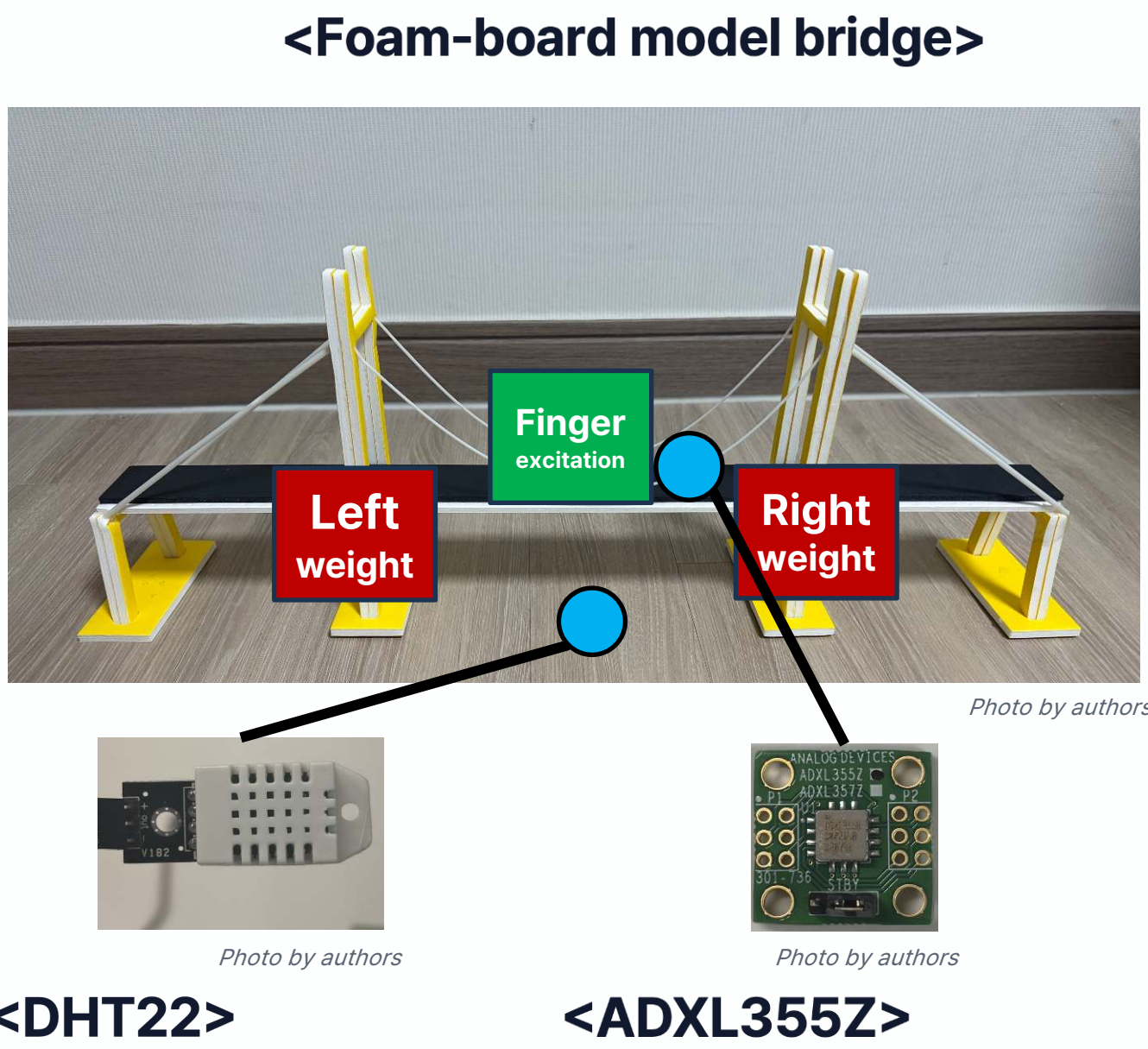
05 Stage 1: Foam-Board Bridge

Method

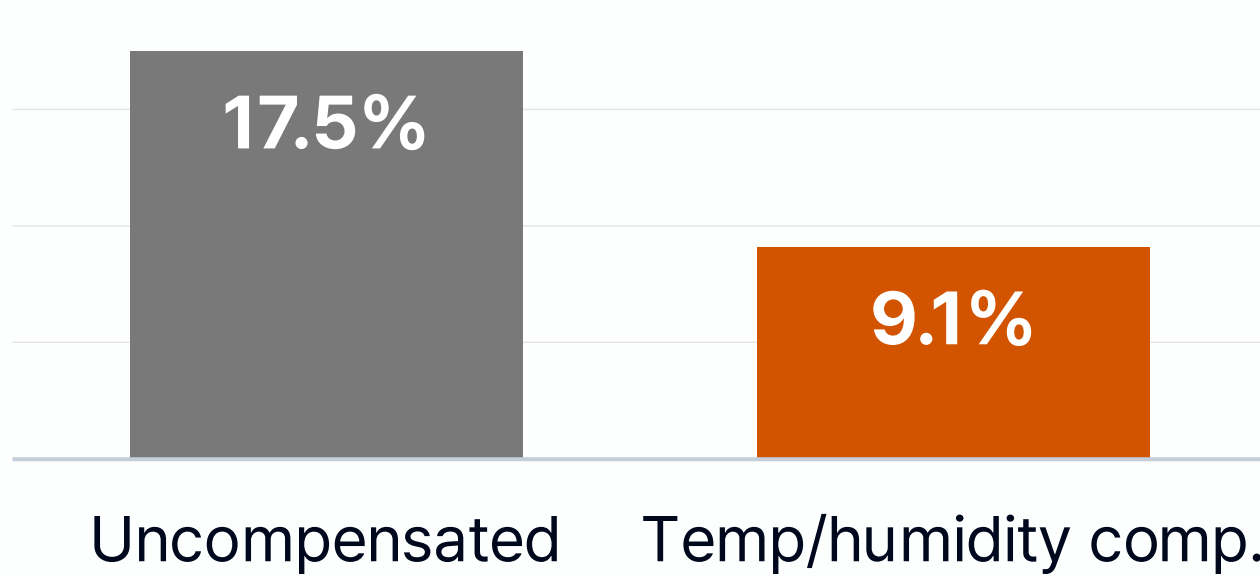
- 1) On the unweighted (normal) model bridge, training data are collected under three conditions: no temp/humidity change, humidity change, and temp + humidity change.
- 2) Weights are then placed left-right as 0-0, 1-0, 1-1 to simulate damage, and test data are collected.

Results

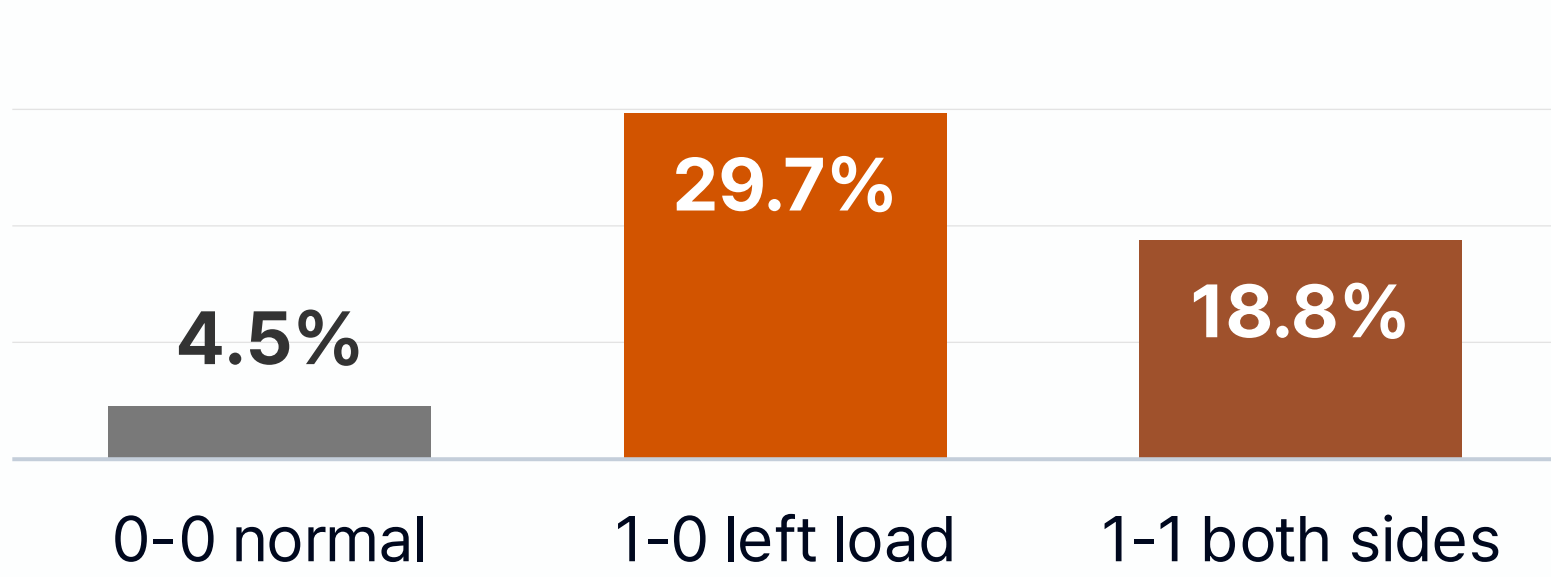
- 1) False-alarm rate 17.5% → 9.1%
- 2) Exceedance rate under simulated damage 4.5% → 29.7%
- 3) Natural frequency shifted by 8.9%



False alarms before/after compensation



Exceedance rate by mass condition



06 openLAB Full-Scale Bridge Analysis

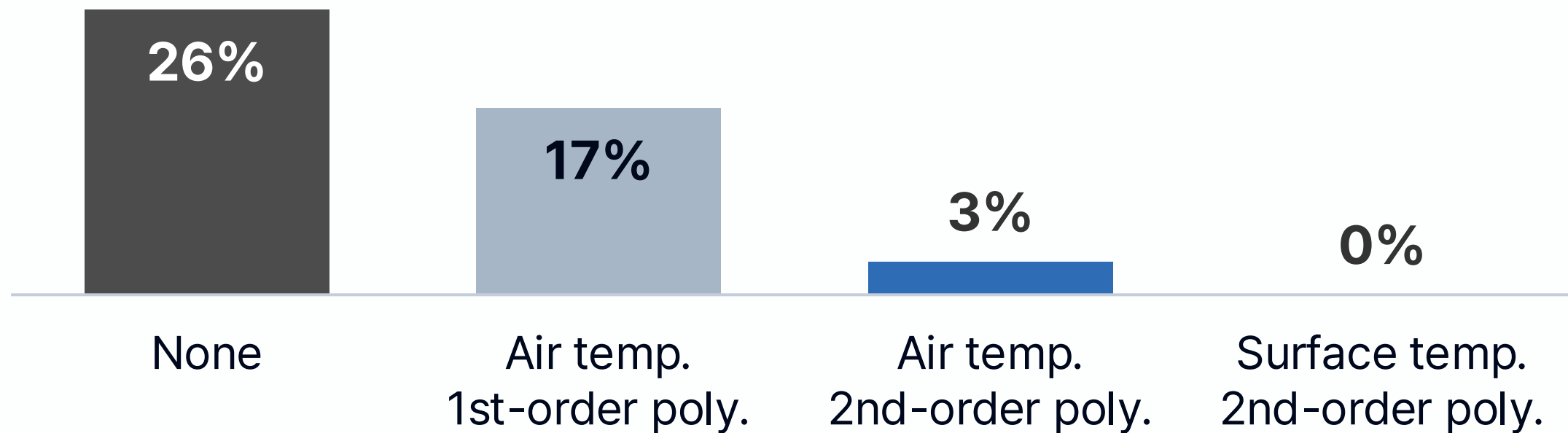
Method

- 1) Learn the normal baseline and threshold from May–June 2024 data of an undamaged real bridge (45 m, 3 spans).
- 2) Add a 7.05 Hz frequency-decision channel (99% range of summer records) to the existing autoencoder, apply it to 69 Sep–Oct records, and compare false-alarm rates by compensation method.



<openLAB full-scale bridge>

False alarms by method (May–Jun → Sep–Oct)

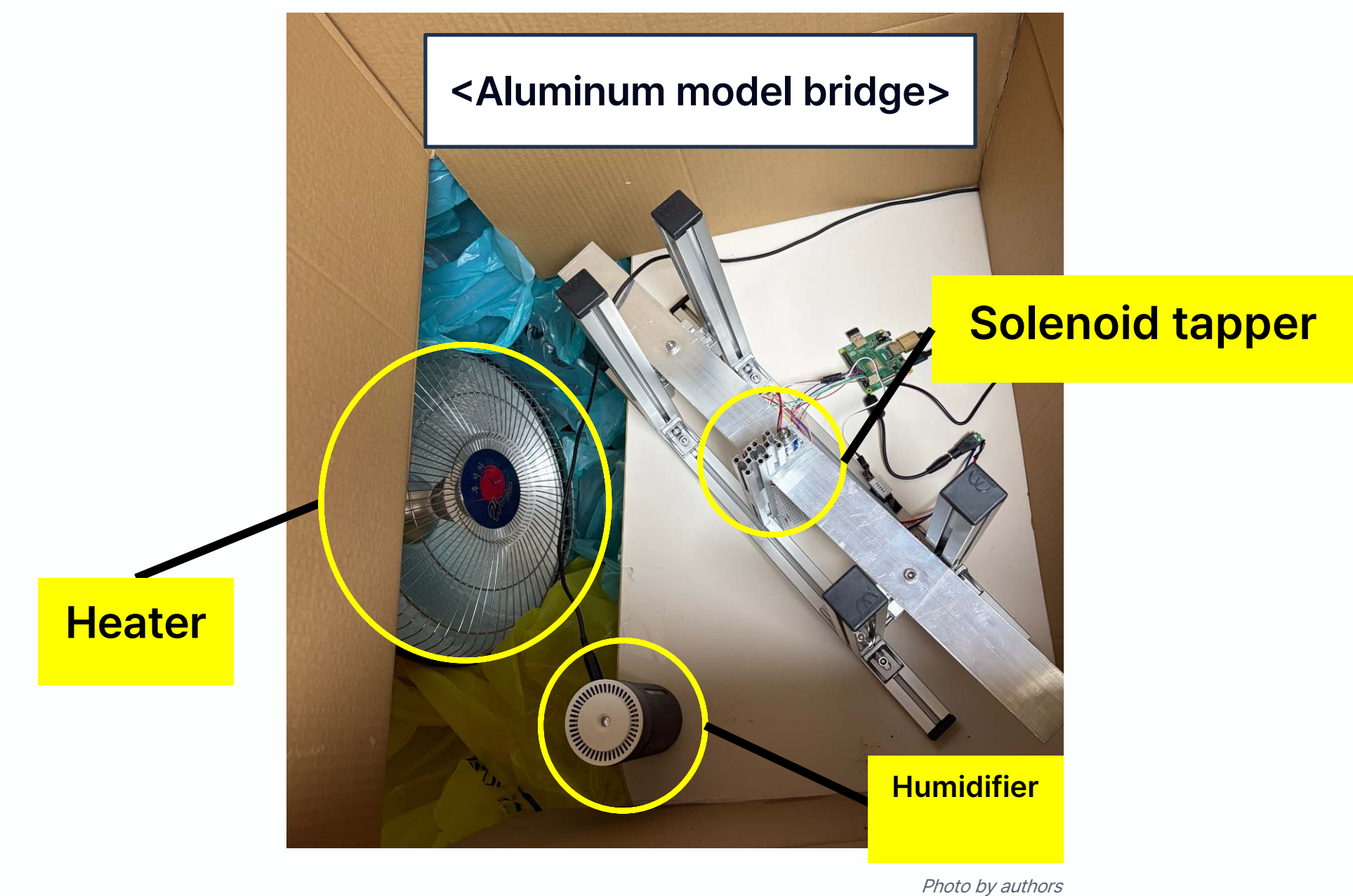


- 1) The existing autoencoder scored 0.1× the threshold even when frequency rose 6–8% in October → blind to seasonal change
- 2) With the frequency channel, false alarms 26% → 0%; long-term effect needs further validation

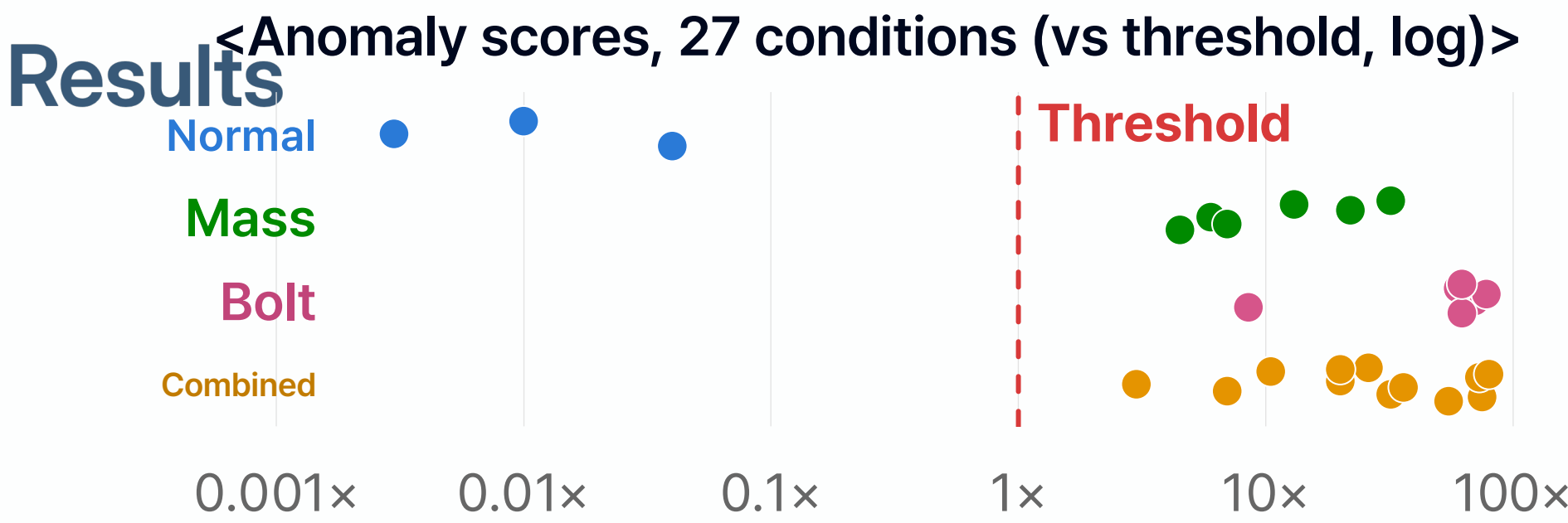
07 Stage 2: Aluminum Model Bridge

Method

- 1) A 27-condition matrix of 3 environment states × 3 mass levels × 3 bolt-loosening levels is collected with automatic solenoid tapping.
- 2) An 8–3–8 Autoencoder is trained on 360 normal windows, the threshold set on a separate 90, and 840 structural-change windows evaluated.



Results



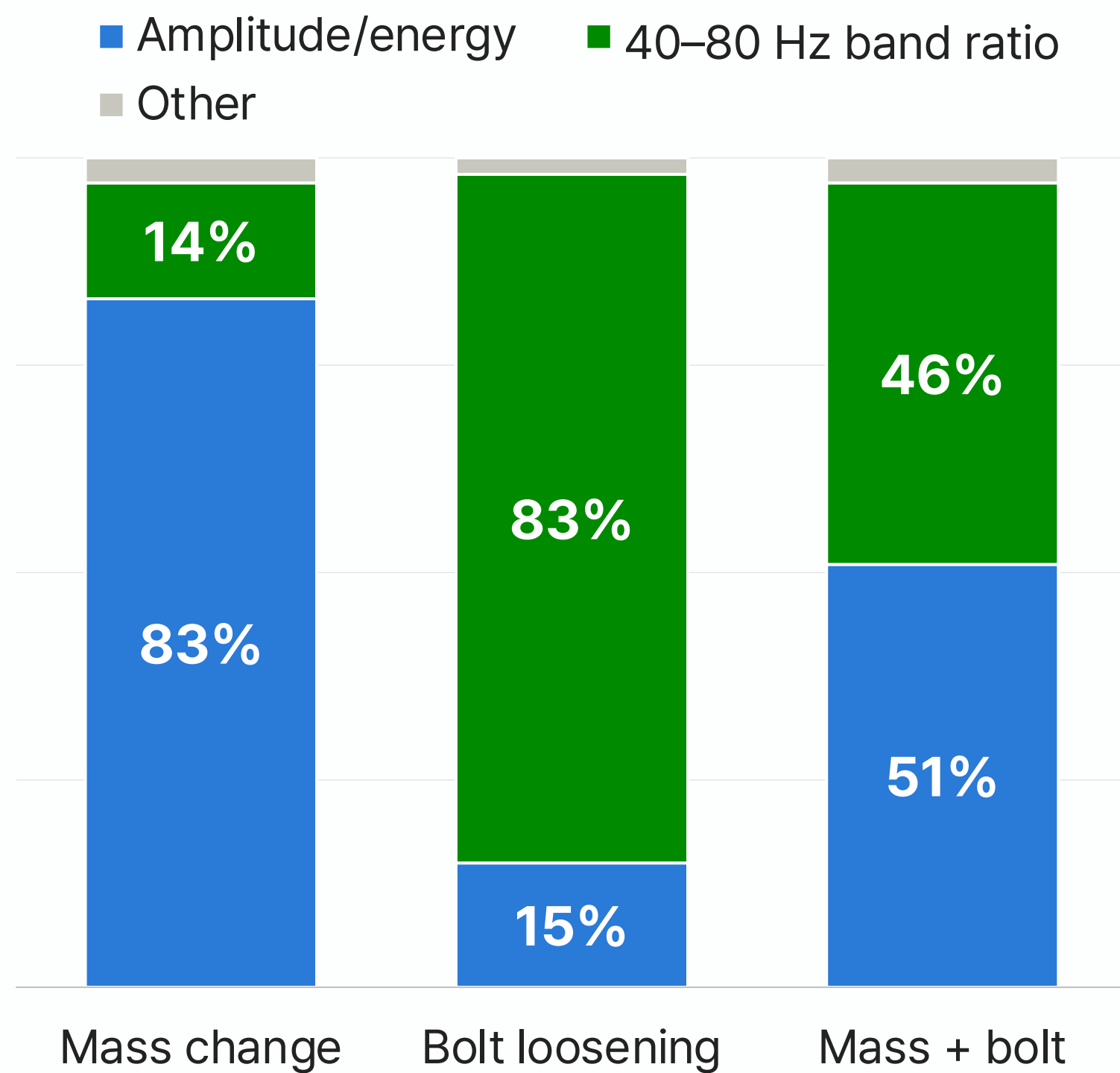
- 1) The temp/humidity-compensated model detected all 840 structural-change windows: recall 100%, normal false-alarm rate 4.4%

<Anomaly score by level of change (× threshold)>

Damage type	Normal	Level 1 (M500·B1)	Level 2 (M1000·B2)
Mass (M)	0.01×	5.8×	22.6×
Bolt (B)	0.01×	47.2×	66.0×

- 2) For both mass and bolts, the ratio grows with the level of change, matching the departure from normal

<Reconstruction-error contributions>



- 3) Bolt-type damage is driven by the 40–80 Hz band ratio (76–90%), mass-type by amplitude/energy (73–80%), so damage types separate and the cause of structural change can be inferred

08 Stage 3: Real Bridges

1. Bridge selection

Bridge	Dominant peak f_p	Peak clarity	Decision
E	1.74 Hz	17,369	Selected
C	3.81 Hz	17,632	Selected
B	3.69 Hz	1,698	—
A	2.62 Hz	194	—
D	No peak	3.3	—

Ambient vibration measured 30 min each; E and C chosen with peaks >10,000× clearer

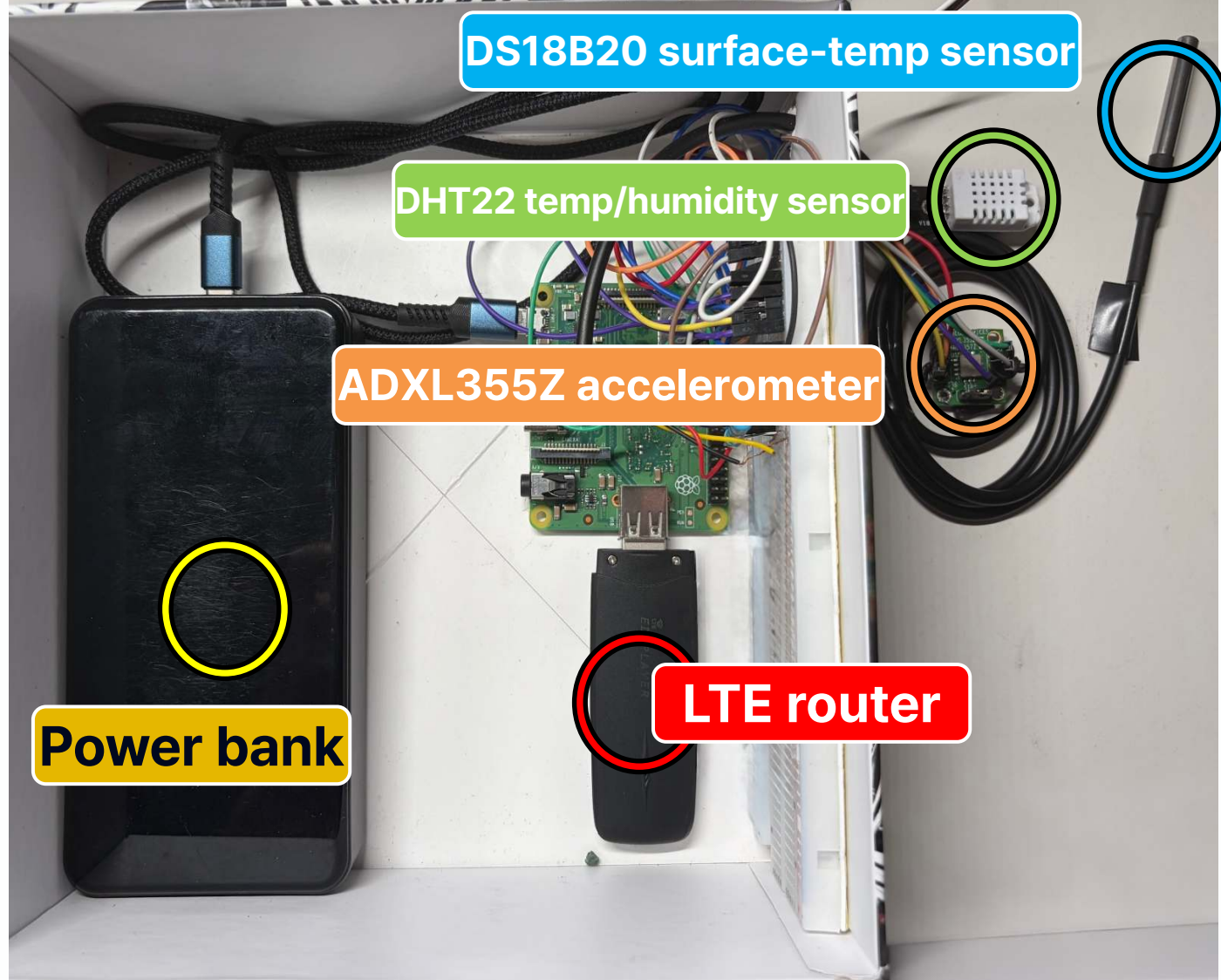
* Peak clarity = dominant-peak PSD ÷ median PSD over 1–20 Hz (Welch, z axis)



<Bridge E (Paju)>

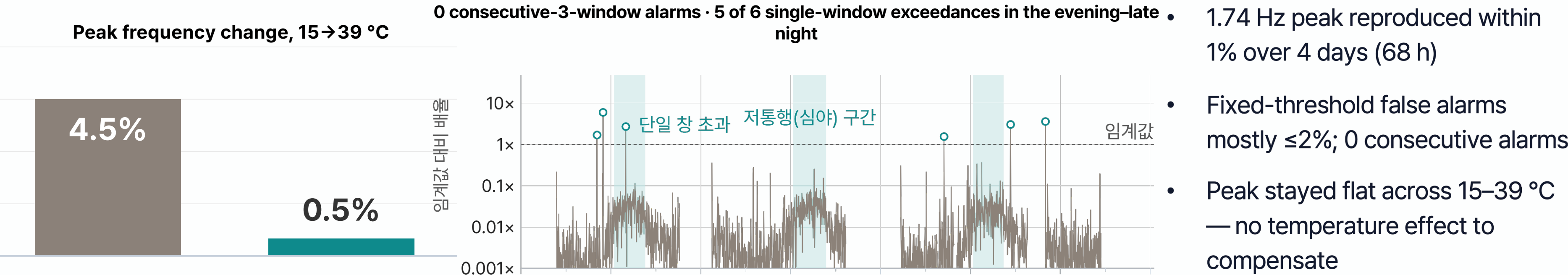


<Bridge C (Changwon)>

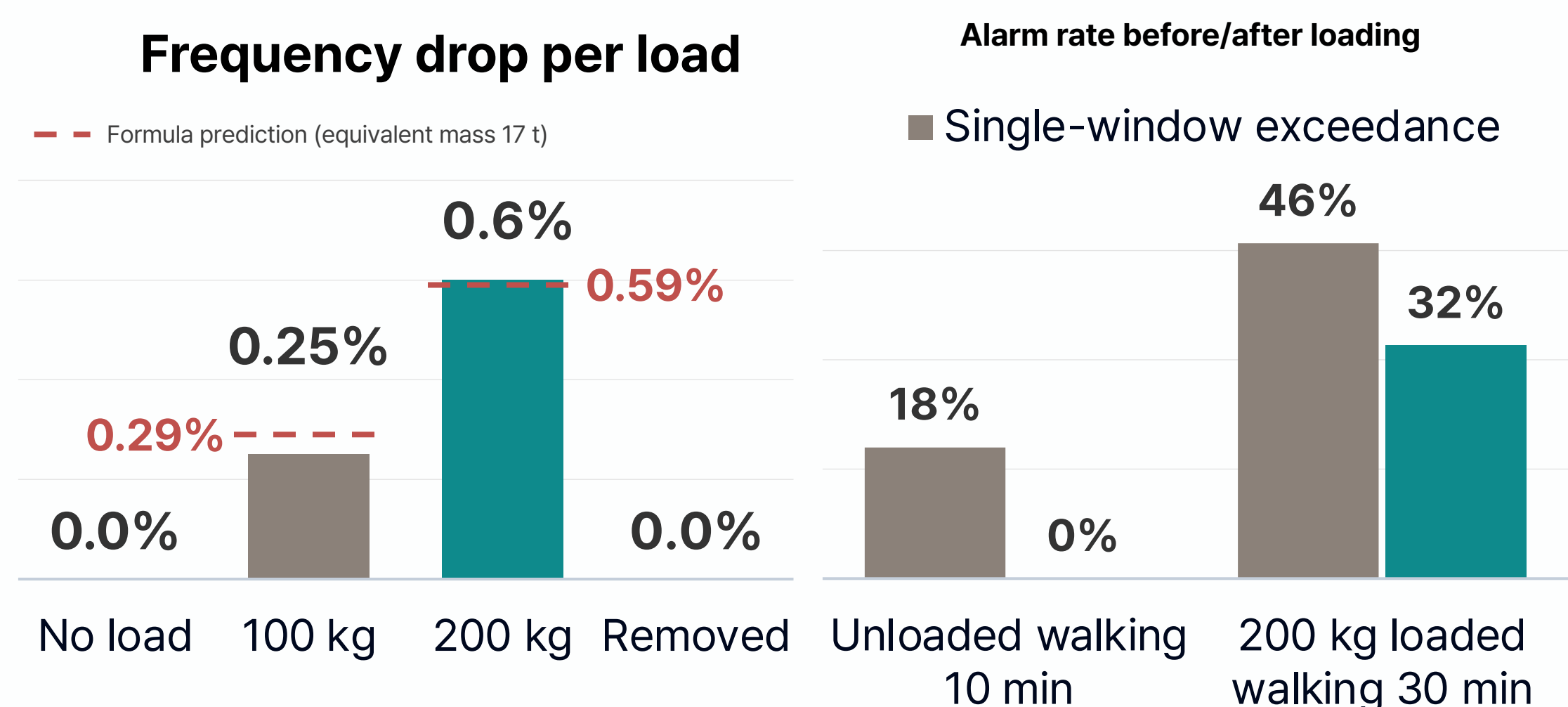


<Stage-3 field node>

2. Bridge E: 4-day measurement



3. Bridge C: detection test



<Weights for detection test>

- Unloaded 10 min: 0 consecutive-3-window alarms
- 200 kg loaded 30 min: 46% of windows above threshold
- Frequency drop matched the added-load prediction

09 Conclusion & Expected Impact

Conclusion

Model bridges (foam board, aluminum)

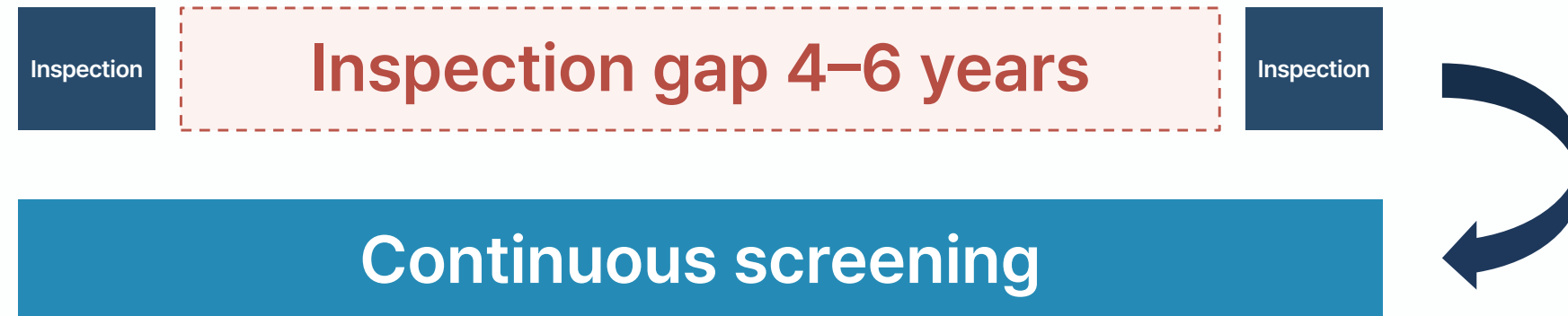
- All 840 windows of 27-condition change detected (recall 100%)
- Contributions separate types and rank departure from normal
- Humidity-segment false alarms 17.5% → 9.1%
- Compensation effect confirmed under large environmental change

Real bridges (overpasses E, C)

- Mode peaks reproduced within 1%
- 200 kg (~1% of equivalent mass) detected by heel-drop frequency drop and recovery
- Early autumn: temperature effect at noise level, no gain from compensation
- False alarms came from weak excitation at low traffic

Expected Impact

⓪ Continuous monitoring fills the gap between inspections



⓪ Monitor 36+ bridges for the same cost

Existing SHM



This system



≈ x36

Future Improvements

- 1) Re-validate compensation with long-term observation including sub-zero periods
- 2) Integrate the frequency-decision channel into the real-time system to watch frequency-type change